

Niroshan Sivarajah

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🌐 LinkedIn | 🏠 GitHub | 📁 Portfolio

📍 Chavakachcheri, Jaffna, Sri Lanka

About Me

Final-year Computer Science and Engineering undergraduate at the University of Moratuwa with a strong interest in backend engineering, distributed systems, and full-stack application development. Skilled in Java, Python, Spring Boot, React, MySQL, Docker, and cloud technologies. Passionate about building practical, scalable, and reliable software solutions while continuously improving as a software engineer.

Education

• University of Moratuwa

BSc Eng Hons in Computer Science and Engineering

May 2022 - June 2026

• J/Chavakachcheri Hindu College

GCE Ordinary Level and GCE Advanced Level

Jan 2012 - Oct 2020

◦ GCE Advanced Level (Physical Science): 3A

◦ GCE Ordinary Level: 8A, 1C

Experience

• Software Engineering Intern - ZeroBeta

Dec 2024 - May 2025

- Designed and developed scalable REST APIs using Java Spring Boot with MySQL integration
- Improved application performance by implementing Redis caching and Kafka-based asynchronous messaging.
- Automated database schema migrations using Liquibase and containerized backend services with Docker.
- Implemented and maintained unit tests using JUnit, Mockito, and Vitest to ensure code reliability and improve test coverage.
- Documented APIs using Swagger and ensured code quality through SonarQube analysis.

Technologies: Spring Boot, Node.js, TypeScript, React, MySQL, Docker, Redis, Kafka, AWS

Tools: Bitbucket, Jira, SonarQube, Postman

Projects

• Tamil-English Code-Switched Speech Recognition System (Group)

Research and Development

Final Year

- Developed a Tamil-English code-switched Automatic Speech Recognition (ASR) system using Transformer-based models.
- Built and curated a 35-hour bilingual speech dataset for model training and evaluation.
- Fine-tuned Whisper models using LoRA and implemented XLM-R and mT5 based post-processing.
- Evaluated model performance using Word Error Rate (WER) and Point of Interest Error Rate (PIER).

• Student Management System (Individual)

Java, Spring Boot, MySQL

Dec 2025

- Designed and developed a role-based Student Management System for managing students, teachers, modules, departments, and enrollments.
- Implemented JWT-based authentication and authorization (access & refresh tokens) using Spring Security with stateless session management and BCrypt password hashing.
- Built RESTful APIs with full CRUD operations, enforcing role-based access control (Admin, Teacher, Student).
- Integrated Spring Data JPA & Hibernate with MySQL, applying DTOs and ModelMapper for clean API-persistence separation within a layered architecture featuring centralized exception handling and CORS configuration.

• Bank Transaction and Loan Processing System : Database Systems (Group)

React, Node.js, MySQL

Semester 03

- Developed a core banking system designed to manage branches, internal fund transfers, and loan processing.
- Contributed to designing and normalizing the database schema, implementing secure user input handling, and developing role-based access controls.

Technical skills

- **Programming Languages:** Java, Python, C++, JavaScript, TypeScript
- **Web & Backend:** React, Node.js, Express.js, Spring Boot, Next.js
- **Databases:** MySQL, PostgreSQL, MongoDB
- **DevOps & Cloud:** Docker, AWS
- **Version Control:** Git, GitHub, Bitbucket
- **Developer Tools:** VS Code, IntelliJ IDEA, WebStorm, Postman, Jira

Honors and Awards

- **Mathematics Olympiad** 2014, 2015
Provincial level participation
- **Maths Quiz** 2013
2nd place in zonal level competition
- **Athletic Meet** 2015, 2016
200m, 400m, 800m, 4x400 relay provincial level participation
- **Student prefect** 2016-2020
J/Chavakachcheri Hindu College

Membership

- **Tamil Literary Association of University of Moratuwa** 2022 - Present

Other skills

Languages: English (fluent) | Tamil (native) | Sinhala (basic)

Soft skills: Quick learner, Teamwork, Communication skills, Problem solving

References

1. Dr. Buddhika Karunaratne

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2. P. Ajanthasiri Gamage

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